

Explainable and Multimodal AFC

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Github : <https://github.com/Azulia888/stage-M1>

1 Introduction

Automatic Fact-Checking (or AFC) is a subject that has garnered a lot of attention through the years, and for good reasons. Fake news and manipulations are becoming more and more widespread, especially with the hegemony of social media. The influx of content is so massive that journalists and fact-checkers cannot report all fake news.

Thus, the idea of an AFC system was born. If there were too many fake news for human minds to tackle, then artificial ones should do it instead. But, while many such systems have been proposed, none of them have yet been fully adopted. The issue does not seem to stem from a simple lack of accuracy, as some models can be very good, or a lack of accessibility, as some simply require asking a question to a free Whatsapp phone number.

We think that this lack of wide-spread adoption can be explained by two major reasons. The first is that most of these models are not as explainable as they should. Journalism is a field where truth matters a lot, and fact-checkers will never be content with citing a single secondary source to make their decisions. Yet, most of the explainability done in the AFC field is limited to linking to an article from a trustworthy fact-checking organism (like Snopes).

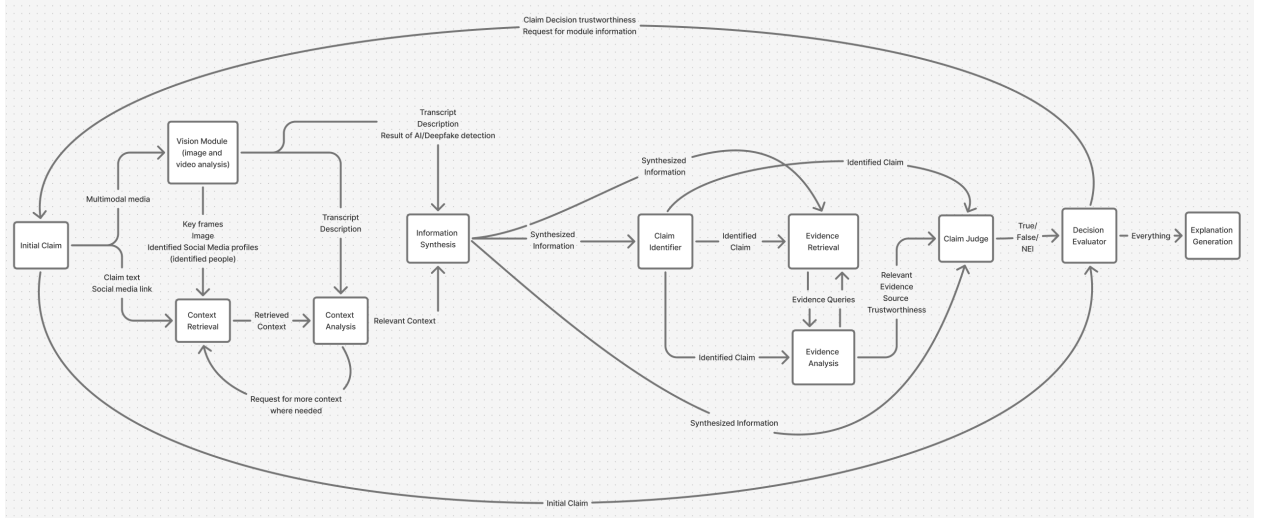
This leads to the second reason. Most, if not all, AFC systems rely on pre-existing fact-checking articles to determine whether the claim is true or false. This seems like a good thing, as it means that the AFC will not hallucinate information, and instead rely on trusted sources. However, it also means that the AFC is relegated to a bridging role between fact-checking organisms and the end user; a sort of specialized search engine.

However, we think there is an overlooked solution to this problem. AFC systems are always made with the goal of replacing journalists. They are made to help a normal person determine if what they are looking at is true or false. Since AFC systems are not meant to propagate more disinformation, this requires them to be perfectly accurate, something which will probably never happen. In a field as important as journalism, an accuracy of 0.95 is not enough.

Built for journalists and fully explainable by design, we propose an AFC system which is meant to help journalists. More than an AI, it can be viewed as a collection of simple tools working together. Tools that journalists already use in their fact-checking process, but automated and explained in a way that is easy to understand. As it is meant to be used by journalists, this collection does not have to always be accurate. It alone will never determine whether something is true or false. The tools used will be described further in the rest of this article, but they are all tools fact-checkers are already familiar with (like Reverse Image Search).

2 Workflow Overview

In this section, we propose an automated workflow that would allow a journalist to simply input a link to a social media post and have it automatically analyzed.



A social media post is first given with its URL and downloaded with yt-dlp. If it contains a video or an image, it is sent to the Vision Module for further analysis. This module is developed in the next section.

Information is then sent to the Context Retrieval module, which will search Internet for relevant information about the given post. This can be made through Reverse Image Search or through social media profile analysis (previous posts, followers, description, etc), to name a few. The raw, retrieved context is then given to the Context Analysis, where a LLM processes it into more digestible data.

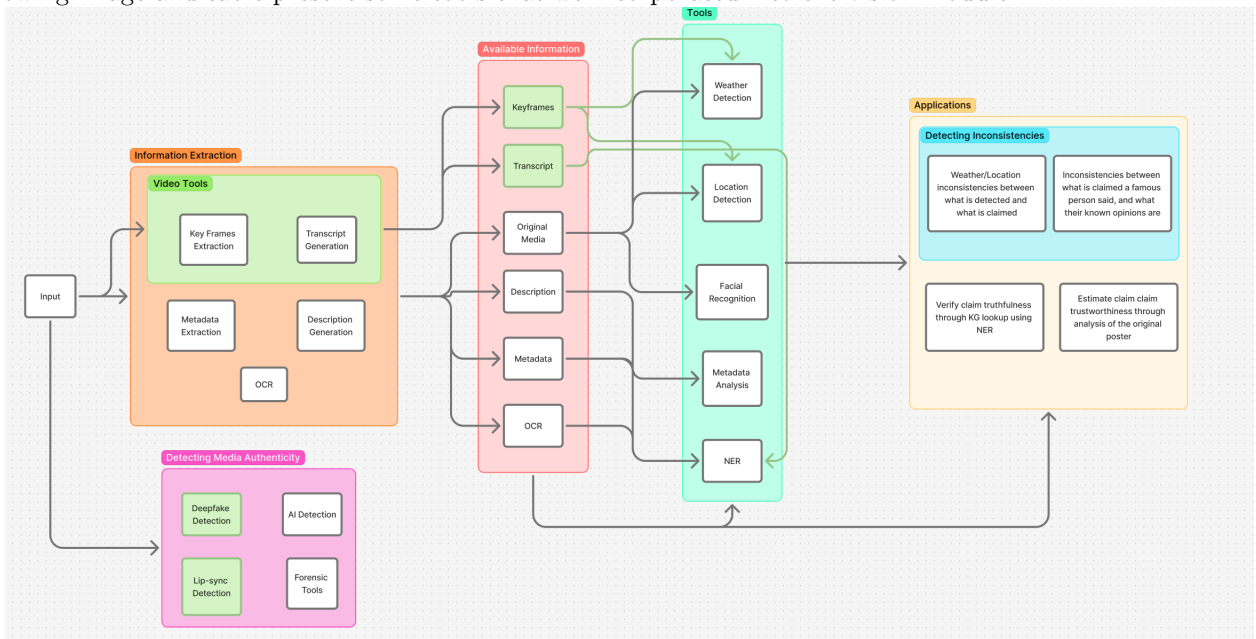
The information found by those two modules is then given to another LLM for synthesis. This synthesis does not destroy the previous information, and it is still available for the user to look at if needed. However, a lot of it will be useless or redundant, and LLMs are useful in cases like this one. Shorter, more concise text will be more useful down the line for other AIs to process.

Once synthesized, the information is given to another LLM, who will try to identify the claims being made by the post. Evidence is then retrieved and analyzed, before the claims are judged as either true or false. If the final LLM decides that there are not enough information to make a good decision, then the process can start over. This is the part which will require the most human intervention, as machines are not good at finding primary sources and they cannot be allowed to ask questions to real people.

Finally, the Explanation Generation module will take the outputs of all the modules and shape them in a way that can be easily understood by the user. This will be described further in its own section.

3 Vision Module

A lot of fact-checking work can be done on a video or an image, and the video or the image itself. The following image and table present some tools that we incorporated into the vision module.



Tool	Data	Why?	How?	FC-task	Corroborating Tools	Required External Evidence	XAI Applications
Keyframe extraction	Video	Used in other image-only tools	Katna; Other techniques				Selected frames should be shown
Transcript	Video	Used in other text-only tools	STT technologies				
Description	Video	Used in other text-only tools	VLM				
OCR	Image	Identifying dates, locations, people; detecting inconsistencies	OCR technologies	OOB detection	Title; Transcript	Information about the identified elements	Directly showing the inconsistencies
Deepfake Detection	Video	Detecting deepfake	Deepfake detection tools	Manipulation Classification			Results + Confidence
AI Detection	Video; Image	Detecting AI-generated media	AI-generated media detection tools	Manipulation Classification			Results + Confidence
Metadata Analysis	Metadata	Can help detect inconsistencies with information from other tools	yt-dlp retrieves metadata from SNS; EXIF	OOB detection	Title; Transcript; Description; OCR	Information about the poster	Inconsistencies detected
Lip-sync Detection	Video; Audio	Detecting out-of-context or manipulated audio	SyncNet; Wav2Lip	Manipulation Classification	Deepfake/AI Detection		Results + Confidence
Weather Detection	Video; Image	Detecting if the weather for the claimed date/location is coherent	VLM; Specialized CNNs	OOB detection	Metadata; Geolocation	Actual weather reports for the claimed date/location	Identified weather and (in)consistencies
Geolocation	Image	Detecting if the identified location is the same as the one claimed	Landmark recognition; GeoSpy	OOB detection	Metadata; Transcript		Identified location (in)consistencies
Facial Recognition	Video; Image	Detecting famous people allows us to detect potential inconsistencies in what is claimed	Specialized model with a famous people database	Manipulation Detection Veracity Classification	Deepfake/AI Detection Title; Transcript	Wikipedia pages of identified people	(In)consistencies between claimed discourse and actual opinions
NER	Transcript OCR Description	Allows for further connections to KG	spaCy	Claim Identification	Transcript; Description	KG related to identified named entities	Found entities and why (+ connections to KG)